

SOMIK GHOSH

Senior Data Engineer | Lead Data Engineer | Principal Data Engineer | Data Architect

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EXECUTIVE SUMMARY

Senior Data Engineering and Data Platform Leader with 15 years of experience designing, modernizing, and scaling distributed data platforms across banking, healthcare, retail, insurance, telecom, and digital media. Specialized in enterprise data lakes, data warehouses, ETL/ELT modernization, Spark, Kafka streaming, Databricks lakehouse migrations, AWS cloud data platforms, Snowflake/Redshift/Teradata ecosystems, governance, data quality, and platform reliability.

Architecture-focused leader for distributed systems, cloud transformation, event-driven platforms, CI/CD automation, Terraform/IaC, and secure governed data consumption across regulated environments.

15+ engineers led
10TB+ daily data
500M+ monthly records
1B+ monthly events
70% faster processing
\$2M+ annual savings

CORE DATA ENGINEERING COMPETENCIES

DISTRIBUTED SYSTEMS & SCALE Spark/PySpark optimization; partitioning; parallel processing; workload tuning; high-throughput ingestion; petabyte-scale architecture; SLA-driven reliability.	DATA LAKES, WAREHOUSES & LAKEHOUSE Databricks lakehouse; enterprise data lakes; Redshift; Snowflake; Teradata; dimensional models; star schemas; analytical serving layers.
ETL, ELT & STREAMING Informatica; Spark ETL; SQL/PLSQL optimization; Airflow; Kafka; Kinesis/MSK; API/SaaS/legacy integration; recovery patterns.	CLOUD & PLATFORM ENGINEERING AWS EMR, Glue, Redshift, Athena, Kinesis, MSK, Aurora, DynamoDB; Terraform; Docker; Kubernetes; GitOps; CI/CD; Azure/GCP familiarity.
GOVERNANCE, QUALITY & COMPLIANCE Data quality gates; lineage; metadata; cataloging; PII controls; GDPR; HIPAA; SOX; auditability; regulated banking and healthcare platforms.	ARCHITECTURE & LEADERSHIP Technical roadmaps; architecture reviews; stakeholder workshops; reusable data assets; team leadership; mentoring; cost optimization; delivery governance.

PROFESSIONAL EXPERIENCE

Engineering Specialist / Staff Engineer	Commonwealth Bank of Australia	Apr 2025 - Present
- Lead architecture and delivery of layered enterprise banking data platforms for risk analytics, regulatory reporting, forecasting, customer insights, and ML-ready data consumption. - Design scalable integration patterns, reusable data assets, governance controls, lineage tracking, and automated quality gates for regulated financial workloads. - Own architecture decisions for reliability, secure data access, zero-downtime deployment, and consistent analytical delivery across banking domains. - Partner with analytics, risk, compliance, engineering, and business stakeholders to convert banking requirements into platform roadmaps and delivery plans.		
Senior Manager - Data Engineering	Tredence Inc.	Nov 2024 - Mar 2025
- Led AWS big data pipelines, increasing processing throughput by 70% through Spark optimization, workload tuning, partitioning, and intelligent resource provisioning. - Migrated legacy ETL workloads to Databricks lakehouse architecture, improving pipeline efficiency by 40% and enabling unified batch, streaming, analytics, and ML feature workloads. - Managed 15 data engineers delivering enterprise BI platforms, feature pipelines, and real-time operational dashboards for Fortune 500 clients. - Implemented GitOps CI/CD, IaC, automated testing, and release controls, reducing deployment lead time from weeks to hours with 99.9% release success.		
Technical Architect Manager	Impetus Technologies	Feb 2023 - Oct 2024
- Acted as principal architect for petabyte-scale data platforms, improving scalability, reliability, query performance, and enterprise analytics availability. - Designed AWS Well-Architected data solutions across ingestion, transformation, serving, analytics, and governance using Kinesis/MSK, Glue/EMR, Athena, QuickSight, and cloud-native storage patterns. - Optimized hybrid data lake and warehouse architectures using partitioning, materialized views, query tuning, and caching strategies to improve latency and workload efficiency. - Established governance standards covering metadata, lineage, PII controls, cataloging, quality, GDPR/SOX compliance, and self-service analytics enablement.		

- Engineered federated ETL across 50+ SaaS, API, legacy database, and operational sources, improving executive decision intelligence and strategic reporting.

Technical Architect / Associate Technical Architect **Carelon Global Solutions / Elevance Health** Apr 2018 - Jan 2023

- Designed HIPAA-compliant healthcare data fabrics integrating claims, clinical, pharmacy, and socio-economic datasets for population health, risk, provider, and value-based care analytics.
- Led IngenioRx PBM analytics architecture for adjudication logic, formulary compliance, rebate modeling, provider network optimization, forecasting, and competitive intelligence workflows.
- Established reusable healthcare data product patterns using microservices, Kafka event streaming, and domain-driven design, reducing time-to-market for new analytics use cases by 60%.
- Delivered ML-enabled personalization and lifecycle analytics capabilities that improved member retention KPIs by 25% through trusted data pipelines.

Senior Consultant - Data Engineering **Deloitte Consulting** Oct 2015 - Apr 2018

- Led ETL and data architecture for Fortune 100 transformation programs, delivering 100+ Informatica workflows processing 10TB+ daily volumes with 99.99% data accuracy.
- Converted executive business requirements into technical blueprints, data models, test automation strategies, and production-grade analytical solutions.
- Mentored 12 junior consultants across data engineering and consulting delivery, contributing to 90% internal promotion within 18 months.

Software Associate **Cognizant Technology Solutions** Jun 2014 - Sep 2015

- Developed Informatica ETL pipelines for telecom billing warehouses, processing 500M+ monthly records using partitioning, incremental loads, recovery, and validation controls.

Technical Analyst **IBT Media** Aug 2011 - May 2014

- Designed star-schema data marts and analytics pipelines for digital media platforms tracking 1B+ monthly impressions across 50+ business KPIs.

EDUCATION & CERTIFICATIONS

- Bachelor of Computer Applications, Techno India University, West Bengal, India | 2011.
- AWS Certified Data Analytics - Specialty | Amazon Web Services Training and Certification | Credential visible via Credly.
- Databricks Academy Accreditation - Generative AI Fundamentals | Databricks Badges / credentials.databricks.com.
- HackerRank Silver Badge - Python | HackerRank.

TOOLS & TECHNOLOGIES

- **Distributed Processing:** Apache Spark, PySpark, Hadoop, Hive, Spark SQL, workload tuning, partitioning, parallel processing, batch pipelines.
- **Data Platforms:** Databricks, Snowflake, Amazon Redshift, Teradata, Data Lakes, Data Warehouses, Lakehouse Architecture, Aurora RDS, DynamoDB.
- **ETL / ELT / Orchestration:** Informatica PowerCenter, Apache Airflow, SQL, PL/SQL, Python, Unix/Linux Shell, incremental loads, recovery patterns, data validation.
- **Streaming & Event-Driven:** Apache Kafka, Kafka Streams, AWS Kinesis, Amazon MSK, event-driven architecture, real-time dashboards, API integration.
- **Cloud & Platform Engineering:** AWS, EMR, Glue, Redshift, Athena, Kinesis, MSK, QuickSight, Terraform, Docker, Kubernetes, Jenkins, GitLab CI/CD, GitOps, CI/CD automation; Azure/GCP data platform familiarity.
- **Governance & Quality:** Data Quality, Data Lineage, Metadata Management, PII Controls, GDPR, HIPAA, SOX, Apache Atlas, Collibra, auditability, data cataloging.
- **Architecture & Leadership:** Architecture Review Boards, Technical Roadmaps, Stakeholder Management, Team Leadership, Agile, SAsE, Mentoring, Cost Optimization, Platform Reliability.